Physics 143a - Quantum Mechanics I

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Today

- Course information
- Begin topic 1: The Schrödinger equation
  - Introduction and origins of quantum mechanics

1 The Schrödinger equation

1.1 Introduction

Question 1.1.1: What do we mean by classical physics?

All of our current understanding of physics is based on two major principles:

1. Quantum mechanics
2. Relativity

Course objective

The objective is to introduce the principles of quantum mechanics together with some applications.

Let us first begin with the principles of classical mechanics. Classical mechanics is based on Newton's laws. It could be used to describe the motion of a system of particles including the propagation of mechanical waves.

Question 1.1.2: What is the notion of a particle in classical physics?

Question 1.1.3: What is the notion of a wave in classical physics?

Consider a Newtonian system containing one particle of mass m and external force that gives it potential energy V. [In this course, and in many books on quantum mechanics, the symbol V is used for potential energy, and potential energy is also simply called potential.] According to Newton's second law of motion, $\vec{F} = m\vec{a}$, or the position $\vec{r}(t)$ of the particle obeys

$$m\frac{d^2\vec{r}}{dt^2} = -\vec{\nabla}V. \tag{1.1.1}$$

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Given the initial location $\vec{r}(0) = \vec{r}_0$ and initial velocity $\vec{v}(0) = \vec{v}_0$ at time $t = 0$, and $V(\vec{r}, t)$, we can determine the exact path $\vec{r}(t)$ of the particle at later time t. With $\vec{r}(t)$ in hand, we can predict the velocity and energy of the particle as well. Because Newtonian mechanics predicts the exact evolution and location of the particle, we say the theory is deterministic. [Remark: In chaotic systems, small changes in initial conditions cause large deviation in subsequent motion. However, in principle, given the exact initial values, classical mechanics predicts the exact path.] In particular, the total energy of the particle is

$$E = \frac{\vec{p}^2}{2m} + V, \tag{1.1.2}$$

where $\vec{p} = m\dot{\vec{r}}$ is the momentum of the particle.

Next, let us consider a wave, and to be specific, the electric field $\vec{E}(\vec{r}, t)$ in an electromagnetic wave in free space obeys the wave equation,

$$\vec{\nabla}^2\vec{E} - \frac{1}{c^2}\frac{\partial^2\vec{E}}{\partial t^2} = 0, \tag{1.1.3}$$

where c is the speed of light. For a monochromatic plane wave of wavelength λ propagating along $\hat{k}$,

$$\vec{E}(\vec{r}, t) = \hat{n}\, E_0 \cos\left(\vec{k}\cdot\vec{r} - \omega t\right), \tag{1.1.4}$$

where $\vec{k} = k\hat{k}$, $k = \frac{2\pi}{\lambda}$, $\omega = kc$, and $\hat{n} \perp \hat{k}$. For propagation along x with the electric field oscillating along y, $\hat{k} = \hat{x}$ and $\hat{n} = \hat{y}$,

$$\vec{E}(x, t) = \hat{y}\, E_0 \cos\left(kx - \omega t\right), \tag{1.1.5}$$

Question 1.1.4: Can we write the $\vec{E}(x, t)$ above in terms of $\hat{y}\, E_0 e^{i(kx-\omega t)}$? Can the electric field be complex?

1.2 Black body radiation and the birth of quantum mechanics

Question 1.2.1: Looking through a small sipping hole on the lid of a coffee tumbler, the color of the interior wall could be (A). black, (B). white, (C). silver, (D). green, (E). brown.

Question 1.2.2: Why do we see different colors, such as a green shirt or a gray wall with the same incident light falling on them?

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An ideal black body is a perfect absorber of electromagnetic radiation. A black body in equilibrium at a given temperature emits all the energy it absorbs, and maintains the same temperature. In the late 19th century, a number of experiments measured the energy per unit volume per unit frequency (or spectral density $u(\omega)$) for radiation from black body in equilibrium. A part of one scheme to make a good realization of a black body radiation experiment is shown in the figure below.

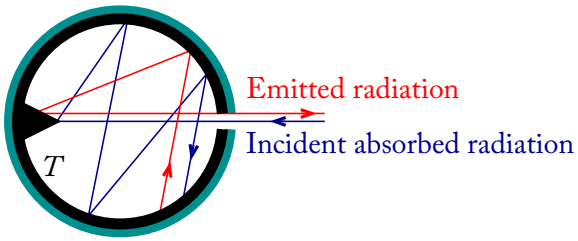


Figure 1.2.1: A black body radiation cavity in equilibrium at temperature T.

A conducting body with hollowed out cavity is covered by an insulating material on the outside. A tungsten incandescent bulb is used as a source of incident radiation that is guided to the black body wall through a small aperture. The incident beam bounces off the interior wall a number of times (since the body is not ideally black) and then gets absorbed. After a while, the black body emits energy at the same rate as it absorbs and maintains an equilibrium temperature. One then makes a histogram of the amount of energy coming out per unit volume of the cavity, called spectral density, through the aperture as a function of frequency. Such experiments in the late 1890's measured spectral densities that look like the schematic plot shown in the figure below.

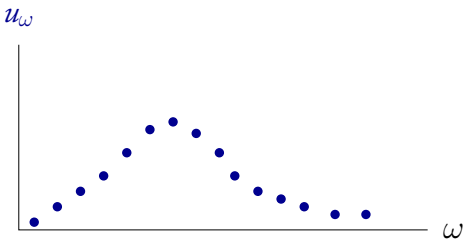


Figure 1.2.2: A schematic plot of experimental data on spectral density at a fixed temperature.

In 1900, Rayleigh derived what is called the Rayleigh-Jeans* formula for the expected spectral

*Jeans corrected the constant coefficient, Rayleigh's counting of states was off by a factor of eight, and completed the derivation in 1905.

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density using classical physics,

$$u_{\text{cl}}(\omega) = \frac{\omega^2}{\pi^2 c^3} k_B T \propto \omega^2, \tag{1.2.1}$$

where $k_\beta = 1.38 \times 10^{-23} J/K = 8.62 \times 10^{-5} eV/K$ is Boltzmann's constant. [The classical formula follows from $u_{\text{cl}}(\omega) = \frac{n(\omega)}{V}(\frac{1}{2}k_B T + \frac{1}{2}k_B T)$, where $\frac{n(\omega)}{V} = \frac{\omega^2}{\pi^2 c^3}$ is the number density (per unit volume per unit frequency) of independent modes of the radiation in the cavity and $\frac{1}{2}k_B T + \frac{1}{2}k_B T$ is the classical energy associated to the two independent quadratic degree of freedom for each mode when the system is in thermal equilibrium at temperature T. In classical statistical mechanics, each quadratic term in the energy contributes an average energy of $\frac{1}{2}k_B T$ for a system in equilibrium at temperature T. It can be shown that the electromagnetic energy density $\frac{1}{2}\epsilon_0 \vec{E}^2 + \frac{\vec{B}^2}{2\mu_0}$ can be simplified into a collection of one-dimensional harmonic oscillators with two quadratic terms for each frequency and polarization.] This is different from the data shown in the figure above. In particular, $u_{\text{cl}}(\omega) \propto \omega^2$ is very different from the data for larger ω. The classical energy over all frequencies per unit volume in the cavity is

$$\frac{U_{\text{cl}}}{V} = \int_0^\infty u_{\text{cl}}(\omega) d\omega = \infty. \tag{1.2.2}$$

Thus, the energy per unit volume diverges because of the ω^2 behavior of $u_{\text{cl}}(\omega)$. This divergence is called "ultraviolet catastrophe."

On October 19, 1900, Planck announced a formula that fits the data. His formula, which is called Planck's black body radiation law, can be written as

$$u_P(\omega) = \frac{\hbar\omega^3}{\pi^2 c^3} \frac{1}{e^{\frac{\hbar\omega}{k_B T}} - 1}, \tag{1.2.3}$$

where $\hbar$ is a constant, which we write in terms of another constant h as $\hbar = \frac{h}{2\pi}$. On December 14, 1900, Planck provided the numerical value of the constant h that fits the data to be $h = 6.55 \times 10^{-34} Js$. Planck's original numerical value of h is remarkably not very far from the current value of $h = 6.626 \times 10^{-34} Js$. The constant h is called Planck's constant and $\hbar$ is called reduced Planck's constant. Furthermore, Planck showed that he could derive the above spectral density if the energy of the oscillating absorbing and emitting atoms in the black body is postulated to come in discrete values (or quanta) in units of $\hbar\omega$,

$$\varepsilon_s = s\hbar\omega, \qquad s = 0, 1, 2... \tag{1.2.4}$$

for each frequency mode ω. Later, in 1905, Einstein proposed that the energy of the radiation

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itself was quantized to explain the photoelectric effect. What this means is that the energy in a radiation at frequency ω comes in and increases as $\hbar\omega$ (one quantum), $2\hbar\omega$ (two quanta), $3\hbar\omega$ (three quanta), etc as schematically shown in the figure below.

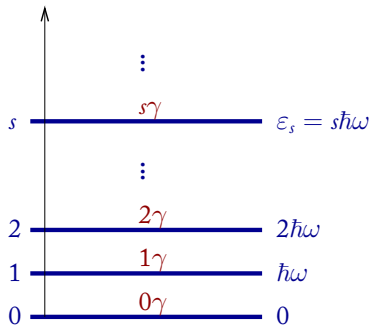


Figure 1.2.3: The quantization of the energy at frequency ω in radiation.

These quanta (or packets of energy) are called photons.[†] Planck's and Einstein's hypothesis amounts to postulating that radiation or light consists of particles, with a particle being a packet of energy. Indeed, later in 1923, Compton's scattering experiment of x-ray radiation off atomic electrons was explained by treating the radiation as composed of particles and the scattering being between two particles, a photon and an electron. Planck's hypothesis of the energy of absorption and emission coming in discrete quanta in fitting the black body radiation experimental data is taken as the origin of quantum mechanics.

Exercise 1.2.1: Determine the energy of an individual photon quantum and estimate the number of photons per unit time in a deep-red laser pointer of wavelength $650nm$ and $1mW$ power.

Solution: The energy of a photon is

$$E = h\nu = \frac{hc}{\lambda} = \frac{6.6\times10^{-34}Js\times3\times10^{8}m/s}{650\times10^{-9}m} = 3.1\times10^{-19}J = 1.9eV. \qquad (1.2.5)$$

$$\frac{N}{t} = \frac{N\hbar\nu}{t\hbar\nu} = \frac{P}{\hbar\nu} = \frac{10^{-3}W}{3.1\times10^{-19}J} \approx 3.3\times10^{15}/s. \qquad (1.2.6)$$

Planck's radiation formula and how it fits the data and how the classical theory fails as ω increases are shown in the schematic figure below.

[†]The word photon was used to describe the quanta of light for the first time in the literature in 1926 by Lewis.

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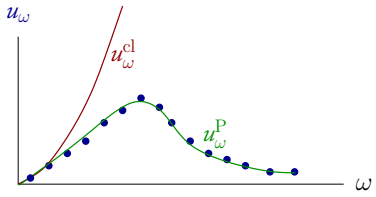


Figure 1.2.4: A schematic plot of how Planck's spectral density $u_\omega^{\rm P}$ fits the experimental data in contrast to the classical spectral density $u_{\rm cl}(\omega)$ which fails significantly as ω increases.

At low ω, $\frac{1}{e^{\frac{\hbar\omega}{k_BT}}-1} \approx \frac{1}{1+\frac{\hbar\omega}{k_BT}-1} = \frac{k_BT}{\hbar\omega}$ and

$$u_{\rm P}(\omega) \to \frac{\hbar\omega^3}{\pi^2c^3}\frac{k_BT}{\hbar\omega} = \frac{\omega^2}{\pi^2c^3}k_BT = u_{\rm cl}(\omega), \tag{1.2.7}$$

and the classical result agrees with the quantum result and data.

Question 1.2.3: Why do Planck's quantum radiation formula and Rayleigh-Jeans's classical formula agree at low frequency.

Exercise 1.2.2: Planck's original black body spectral density formula was written for energy per unit volume per unit frequency ν, where $\omega = 2\pi\nu$, and Planck's constant h rather than $\hbar$. Rewrite $u_P(\omega)$ for $u_P(\nu)$ in terms of ν and h.

Solution:

$$u_P(\nu) = \frac{dU}{Vd\nu} = \frac{2\pi dU}{Vd\omega}\Big|_{\omega=2\pi\nu,\,\hbar=\frac{h}{2\pi}} = \frac{8\pi h\nu^3}{c^3}\frac{1}{e^{\frac{h\nu}{k_BT}}-1}. \tag{1.2.8}$$

Exercise 1.2.3: Rewrite the spectral density $u_P(\lambda)$ for energy per unit volume per unit wavelength rather than frequency.

Solution:

$$u_P(\lambda) = \frac{dU}{Vd\lambda} = \frac{dU}{Vd\nu}\Big|_{\nu=\frac{c}{\lambda}}\frac{d\nu}{d\lambda} = \frac{8\pi h\nu^3}{c^3}\frac{1}{e^{\frac{h\nu}{k_BT}}-1}\Big|_{\nu=\frac{c}{\lambda}}\left(-\frac{c}{\lambda^2}\right) = \frac{8\pi hc}{\lambda^5}\frac{1}{e^{\frac{hc}{k_BT\lambda}}-1}, \tag{1.2.9}$$

where the minus sign in $\frac{d\nu}{d\lambda} = -\frac{c}{\lambda^2}$ just indicates that an increase of energy with increasing ν corresponds to a decrease with increasing λ, and the negative sign is ignored.

Exercise 1.2.4:

The peak frequency $\omega_{\rm peak}$ of the radiation from a black body follows from Planck's

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spectral density $u(\omega) = \frac{\hbar\omega^3}{\pi^2 c^3} \frac{1}{e^{\frac{\hbar\omega}{k_\beta T}} - 1}$ and $\frac{du(\omega)}{d\omega} = 0$. It is given by $\omega_{\text{peak}} = \frac{2.82 k_\beta T}{\hbar}$.

1. Assuming that the radiation from the sun can be roughly modelled as that from a black-body emitter at a surface temperature of 5780 K, determine the peak frequency of the emitted radiation. What is the corresponding wavelength? In which range of the electromagnetic wave spectrum does it belong?
2. Taking the surface of the earth as a black-body emitter at temperature 288 K, determine the peak frequency of the emitted radiation. In which range of the electromagnetic wave spectrum does it belong?
3. The cosmic microwave background radiation (left over from the early universe) has a black-body spectrum at a temperature of 2.73 K. What is the peak frequency and in which range of the electromagnetic wave spectrum does it belong?

1.3 Diffraction and interference of light

Consider a monochromatic laser beam of wavelength λ incident toward a slit of width $a >> \lambda$. The beam passes through the slit pretty much straight and hits the viewing screen as shown in the figure below. The small width shown is to indicate that any measuring device cannot locate the intensity with exact precision.

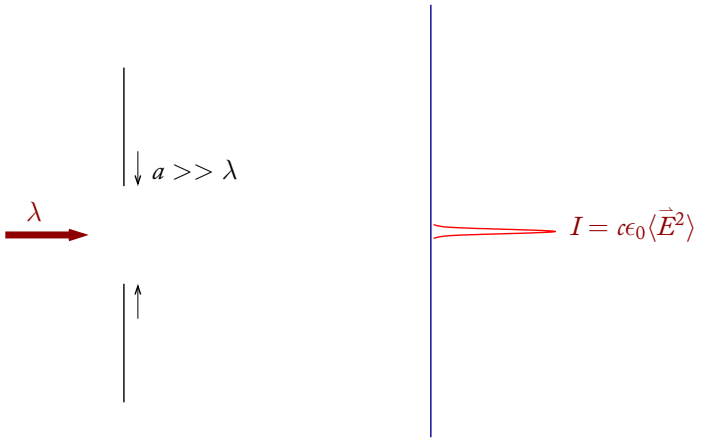


Figure 1.3.1: A laser beam passing through a large slit.

Next, suppose the slit has a width that is comparable to the wavelength of the beam, $a \sim \lambda$. In this case, the resulting beam gets diffracted as shown in the figure below.

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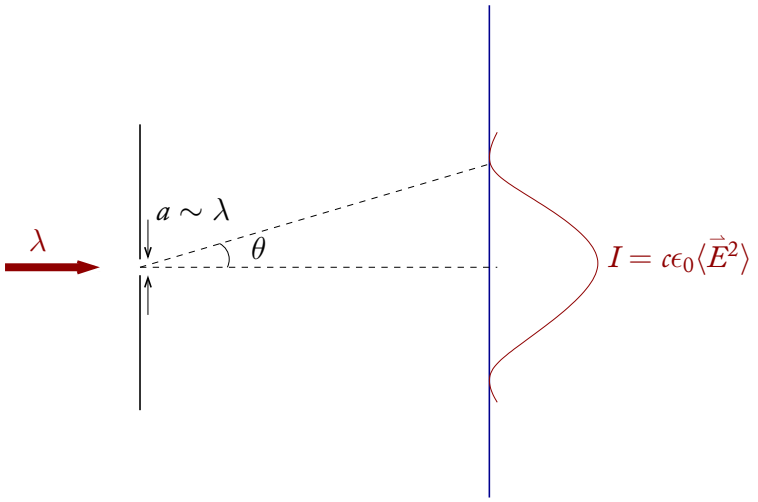


Figure 1.3.2: Diffraction of a laser beam passing through a slit of width $a \sim \lambda$.

The angle θ for the first minimum is given by

$$\sin\theta = \frac{\lambda}{a}, \tag{1.3.1}$$

and $\theta \to 0$ when $\frac{\lambda}{a} \to 0$.

Now consider two slits each of width $a \sim \lambda$ next to each other. First, we open only one slit at a time, and then we open both. The resulting observations are shown in the figure below.

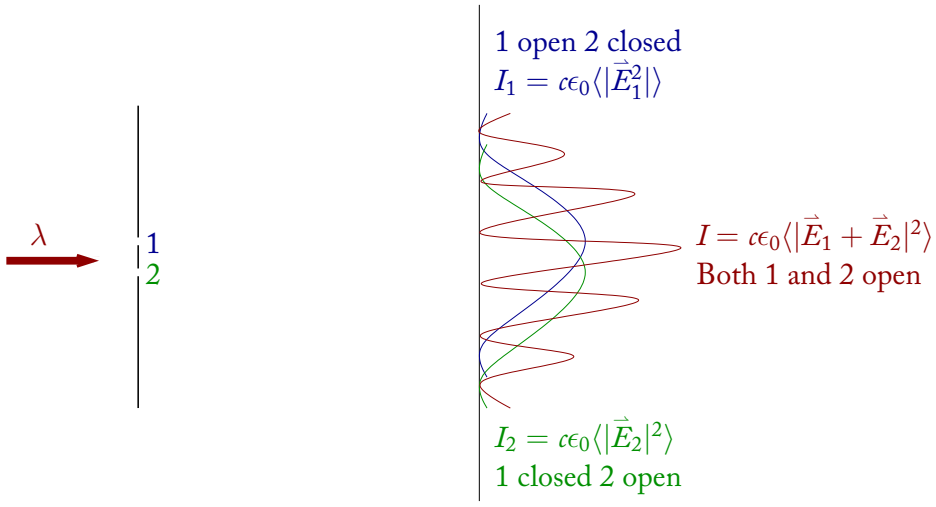


Figure 1.3.3: Diffraction and interference of a laser beam passing through two slits of width $a \sim \lambda$.

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The above diffraction and interference properties demonstrate the wave nature of light. The experiment of double slit interference and the demonstration of the wave nature of light was first done by Young in 1801.

1.4 Diffraction and interference of electrons

View the simulation at

https://physicsworld.com/a/feynmans-double-slit-experiment-gets-a-makeover/.

The paper that reported the results in the simulation is available at

https://iopscience.iop.org/article/10.1088/1367-2630/15/3/033018.

The simulation shown is an accelerated version of an experiment done over two hours with one electron fired per second. An interesting feature of this experiment is the fact that it involved firing one electron per second. There were other previous double-slit electron interference experiments, including one in 1961, but they involved larger intensity electron beams.

Question 1.4.1: What does the simulation demonstrate?

We also know that electrons have point-like mass, charge, and undergo collisions, thus, exhibiting particle property.

Based on all the above observations, we conclude "classical particles" such as electrons exhibit both particle and wave properties and "classical waves" exhibit both wave and particle properties. This phenomenon is called wave-particle duality.

The electron diffraction and interference experiment raises a number of questions.

Question 1: We expect the electrons must have wavelength comparable to the slit width. Because the electrons are fired one at a time and the diffraction and interference pattern survives, we conclude that individual electrons exhibit wave property. But what is the wavelength of an electron?

Question 2: In the case of Young's double-slit interference of light, the intensity of the light at a point on the viewing screen is described by the electric field at that point, and the electric field evolves according to the wave equation. What is the analogue of the electric field in the electron diffraction and interference, and how does it evolve in time (or what is its equation of motion)?

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Question 3: The electron diffraction experiment was done by firing one electron per second. Let us focus on firing only one electron and stopping. Can we predict where on the screen it falls?

Classical physics does not provide answer to these questions. The principles of quantum mechanics answer them as follows.

1.5 Answer to question 1: de Broglie's hypothesis

First, consider light of wavelength λ. Following Planck's hypothesis, $E = h\nu = h\frac{c}{\lambda}$, its wavelength is given by

$$\lambda = \frac{hc}{E}. \tag{1.5.1}$$

On the other hand, following Einstein, the energy of a particle of rest mass m_0 and momentum p is given by $E = \sqrt{p^2c^2 + m_0^2c^4}$. Because light propagates with the speed c in free space, the photons of Planck's hypothesis must be massless, $m_0 = 0$. [Recall that, otherwise, $E = \frac{m_0c^2}{\sqrt{1-v^2/c^2}}$ would become infinity for $v = c$ and $m_0 \neq 0$. Furthermore, light does not have a rest frame.] Therefore, for photons,

$$E = pc. \tag{1.5.2}$$

Combining Planck's and Einstein's formulae, the wavelength corresponding to a photon of momentum p is

$$\lambda = \frac{h}{p}. \tag{1.5.3}$$

In 1923, de Broglie hypothesized the formula $\lambda = \frac{h}{p}$ as a formula that applies to massive particles such as the electron as well. This hypothesis was later first confirmed by the Davisson–Germer experiments involving electron diffraction off crystals in 1923 - 27. The diffraction and interference pattern of the experiment shown in the simulation above was done in 2012 and reported in 2013, and it is also reproduced by treating the electrons as waves with wavelength given by the de Broglie hypothesis.

> **Exercise 1.5.1:** In the electron diffraction experiment shown in the simulation above, electrons are accelerated across a potential difference of $= 600\,V$ each slit has width of $a = 62\,nm$, and the distance between the two slits is $d = 272\,nm$. Determine the de Broglie wavelength of an electron, compare it with a, and discuss what it means. For contrast, determine the de Broglie wavelength of a macroscopic object of mass $1\,kg$ moving with velocity of $1\,m/s$ and discuss whether it exhibits

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wave property in practice. What about an electron moving with a speed of $10^5 m$ in a hydrogen atom?

Solution: The kinetic energy of an electron is $E = e\Delta V = 600\ V = 1.6\times10^{-19} C\times 600V = 9.6\times10^{-17} J$. Its momentum is $p = \sqrt{2mE} = \sqrt{2\times9.1\times10^{-31} kg\times9.6\times10^{-17} J} = 1.3\times10^{-23} kg.m/s$. Its de Broglie wavelength is $\lambda = \frac{h}{p} = \frac{6.6\times10^{-34} Js}{1.3\times10^{-23} kg.m/s} = 5.0\times10^{-11} m = 0.05\ nm$. The angular width of the central diffraction peak is $2\theta \approx 2\sin\theta = 2\frac{\lambda}{a} = 2\frac{0.05\ nm}{62\ nm} = 1.6\ rad$, since $\sin\theta \approx \theta$ for small angle. Therefore, the electrons are diffracted by this angle. The viewing screen in the experiment is located at $D = 30.5\ cm$ from the slits, and therefore, the spatial width of the peak on the screen is $\Delta y = 2\theta D \approx 48\ mm$. The simulation shows the screen magnified by a lens. The angle between consecutive bright or dark fringes is $\theta \approx \frac{\lambda}{d} = \frac{0.05\ nm}{272\ nm} = 1.8\times10^{-4}\ rad$.

For the macroscopic object, $\lambda = \frac{h}{mv} = \frac{6.6\times10^{-34} Js}{1kg\times1m/s} = 6.6\times10^{-34} m$. In order for the object to undergo diffraction, it needs to pass through a slit of similar width, which is impossible for an object of much bigger macroscopic size. Because λ is much smaller than the size of the object, or size of the world in which the object lives in, the wave property of the object is completely negligible and it behaves as a classical object.

For the given electron in a hydrogen atom, $\lambda = \frac{h}{mv} = \frac{6.6\times10^{-34} Js}{9.1\times10^{-31} kg\times10^5 m/s} = 7.3\times10^{-9} m$, which is of the same order as the size of the atom. Therefore, the wave property of the electron is important.

Exercise 1.5.2: Consider the Bohr model of the hydrogen atom with the proton fixed at the origin and the electron moving in a circular orbit around the proton subject to only the Coulomb force.

1. Show that for an electron undergoing centripetal motion of radius r, the total energy of the electron can be written as $E = -\frac{p^2}{2m_e} = -\frac{e^2}{8\pi\epsilon_0 r}$, where ϵ_0 is permittivity of free space, p is the momentum, m_e is the mass, $-e$ is the charge of the electron.

2. Assuming that an integral multiple of de Broglie wavelengths, $n\lambda$, fit into the circumference of the circular orbit, determine the quantized radii of the orbits r_n and the quantized energies E_n as a function of integer n. Write the numerical values in m for r_1 and in eV for E_1.

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3. Write the numerical value of the de Broglie wavelength for $n = 1$ and point out in one sentence what it means that it is comparable to the size of the atom.

4. Determine the ratio of the speed of the electron in the nth orbit to the speed of light, and write the numerical value for $n = 1$. Do you think relativistic effects would be important?

5. Determine the angular momentum of the electron in the nth orbit and show that it is an integral multiple of $\hbar$.

6. According to classical electromagnetism, an accelerating charged particle radiates and loses energy at the rate given by the Larmor formula $\frac{dE}{dt} = -\frac{q^2a^2}{6\pi\epsilon_0 c^3}$, where q is the charge and a is the acceleration of the particle. If the electron initially in the $n = 1$ orbit were to radiate at the rate given by the Larmor formula, determine how long it would take for it to spiral and fall into the proton? What does this tell us about Bohr's model of the hydrogen atom with the electron orbiting around the proton? [You may take the radius of the proton to be negligible compared to the radius of the $n = 1$ orbit.]

Question 1.5.1: What would happen if Planck's constant were $\hbar = 1\,Js$?.

1.6 Answer to question 2: Wavefunction and the Schrödinger equation

According to the principles of quantum mechanics, the analogue of the electric field that describes the interference pattern on the screen is called the wavefunction $\psi(\vec{r}, t)$.

The wavefunction for a particle of mass m with potential energy $V(\vec{r}, t)$ evolves according to the Schrödinger equation,

$$i\hbar\frac{\partial\psi(\vec{r}, t)}{\partial t} = -\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r}, t) + V(\vec{r}, t)\psi(\vec{r}, t), \tag{1.6.1}$$

where

$$\vec{\nabla}^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \tag{1.6.2}$$

in cartesian coordinates, $\hbar$ is the same reduced Planck's constant (which we will often simply call Planck's constant) and $i = \sqrt{-1}$.

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When slit 1 is open and 2 is closed, the density of electrons in the diffraction pattern on the screen is described by $\mathcal{P}_1 = |\psi_1(\vec{r},t)|^2$, where $\psi_1(\vec{r},t)$ represents the wavefunction due to the propagation of the electron through slit 1. When slit 2 is open and 1 is closed, it is described by $\mathcal{P}_2 = |\psi_2(\vec{r},t)|^2$, where $\psi_2(\vec{r},t)$ represents the wavefunction coming through slit 2. When both slits are open, the density is given by $\mathcal{P} = |\psi_1(\vec{r},t) + \psi_1(\vec{r},t)|^2$, and the addition of the two leads to the interference pattern seen on the screen. These are schematically shown in the figure below.

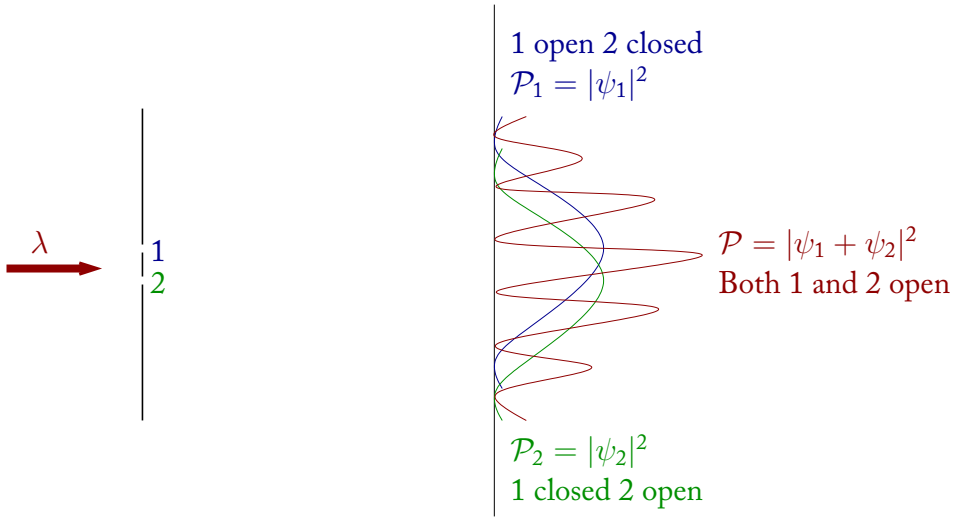


Figure 1.6.1: Density for a single electron in double slits electron diffraction before measurement.

We make three important observations. First, the wavefunction follows a deterministic equation. Given the potential $V(\vec{r},t)$, the wavefunction $\psi(\vec{r},0)$ at some initial time $t = 0$, and appropriate boundary conditions for its spatial distribution, we can determine $\psi(\vec{r},t)$. In other words, the wavefunction follows a deterministic equation. Second, the Schrödinger equation involves $i = \sqrt{-1}$, and the wavefunction is complex, even if we start with a real initial wavefunction. Third, the equation involves first derivative in time and second derivatives in space. For contrast, the wave equation involves second derivatives in both space and time.

Question 1.6.1: Point out why the Schrödinger equation cannot be an equation for a relativistic system.

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1.7 Answer to question 3 part I: Statistical interpretation and Born's hypothesis

According to the principles of quantum mechanics, we cannot predict with certainty where the electron falls on the screen. We can only talk about the probability that the electron falls at a given point on the screen. The probability density that the electron falls at point $\vec{r}$ at time t on the screen is given by

$$\mathcal{P}(\vec{r}, t) = |\psi(\vec{r}, t)|^2 = \psi(\vec{r}, t)^* \psi(\vec{r}, t), \tag{1.7.1}$$

where $\psi(\vec{r}, t)$ is the wavefunction. With this, the electron density pattern seen in the figure above becomes the probability density that a single electron falls at that point, where $\psi(\vec{r}, t)$ is the wavefunction for a single electron. The wavefunction is also often called the probability amplitude or the state (or the state vector).

We summarize Born's interpretation and requirement on the wavefunction as follows.

1. Statistical interpretation: The wavefunction is the probability amplitude and $|\psi(\vec{r}, t)|^2$ is the probability density for finding the particle at point $\vec{r}$ at time t.

2. Normalization: Because the particle must be found at some point at a given time, the total probability for finding the particle over all space must be one,

$$\int_{\text{over all space}} |\psi(\vec{r}, t)|^2 d^3\vec{r} = 1. \tag{1.7.2}$$

   The wavefunction must be square-integrable.

   In particular, it must have finite values everywhere, so that the $\int |\psi(\vec{r}, t)|^2 d^3\vec{r} = 1$. If a wavefunction $\psi_s(\vec{r}, t)$ is obtained as a solution of the Schrödinger equation, it is finite everywhere, and $\int_{-\infty}^{\infty} |\psi_s(\vec{r}, t)|^2 dx = A \neq 1$, then we normalize (or rescale) the wavefunction and promote it to $\psi(\vec{r}, t) = \frac{\psi_s(\vec{r}, t)}{\sqrt{A}}$ so that $\int |\psi(\vec{r}, t)|^2 d^3\vec{r} = 1$ and the wavefunction is normalized. If we find solutions to the Schrödinger equation that are not normalizable, we discard them. However, we will find what are called improper wavefunctions that are not normalizable in the standard sense defined above, but that are useful as we will discuss later in topic 3.

   Question 1.7.1: What is the dimension (or unit) of the wavefunction in three dimensions? What about in one dimension?

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